Clinical Migraine Analysis & Multi-Class Classification Pipeline

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Abstract

This report details an end-to-end exploratory data analysis and machine learning pipeline built on clinical migraine records ($N = 400$). By coupling SQL data wrangling with exploratory visual diagnostics and gradient boosting algorithms (XGBoost), we investigate the non-linear relationships among clinical features and build a predictive diagnostic classifier achieving 93% accuracy.

1 Introduction & Data Integrity

The dataset consists of 24 clinical features tracking patient age, attack duration, frequency, intensity, and associated neurological symptoms. Initial data auditing via SQL queries confirmed a pristine dataset with **zero missing values** and realistic clinical boundaries across all parameters.

2 Exploratory Data Analysis Insights

- **Demographic Concentration:** Age distribution plots confirm that migraine occurrences peak sharply between the ages of 20 and 40, tapering off in later decades.
- **Feature Independence:** Correlation matrix analysis revealed coefficients hovering near zero (maximum magnitude ≈ 0.15), confirming that numerical variables operate independently and reflect non-linear human health patterns rather than rigid linear scalings.

3 Machine Learning Classification

To predict specific migraine classifications, the feature set was preprocessed and trained using an **XGBoost Classifier**.

- **Accuracy:** Achieved an overall test accuracy of **92.5%**.
- **Evaluation:** Multi-class evaluation via confusion matrix heatmaps demonstrated robust diagonal strength for high-support classes (e.g., *Typical aura with migraine*), while highlighting the real-world impact of data scarcity on rare sub-types.

4 Conclusion

This project establishes a robust portfolio template demonstrating competency in SQL querying, exploratory clinical storytelling, and production-grade machine learning evaluation. Future iterations will focus on addressing class imbalance via SMOTE and deploying interactive web interfaces.